BCS-066	SOFTWARE PROJECT MANAGEMENT	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To develop awareness regarding the theoretical and methodological issues related to software project management.
CO2	To develop software projects based on current technologies.
CO3	To make them understand the concepts of Project Management for planning to execution of projects.
CO4	To make them understand the feasibility analysis in Project Management and network analysis tools for cost and time estimation.
CO5	To enable them to comprehend the fundamentals of Contract Administration, Costing and Budgeting.

Detailed Syllabus

MODULE I

Introduction and Software Project Planning

Fundamentals of Software Project Management (SPM), Need Identification, Vision and Scope document, Project Management Cycle, SPM Objectives, Management Spectrum, SPM Framework, Software Project Planning, Planning Objectives, Project Plan, Types of project plan, Structure of a Software Project Management Plan, Software project estimation, Estimation methods, Estimation models, Decision process.

Project Organization and Scheduling

Project Elements, Work Breakdown Structure (WBS), Types of WBS, Functions, Activities and Tasks, Project Life Cycle and Product Life Cycle, Ways to Organize Personnel, Project schedule, Scheduling Objectives, Building the project schedule, Scheduling terminology and techniques, Network Diagrams: PERT, CPM, Bar Charts: Milestone Charts, Gantt Charts.

MODULE II

Project Monitoring and Control

Dimensions of Project Monitoring & Control, Earned Value Analysis, Earned Value Indicators: Budgeted Cost for Work Scheduled (BCWS), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Interpretation of Earned Value Indicators, Error Tracking, Software Reviews, Types of Review: Inspections, Desk checks, Walkthroughs, Code Reviews, Pair Programming.

Software Quality Assurance and Testing

Testing Objectives, Testing Principles, Test Plans, Test Cases, Types of Testing, Levels of Testing, Test Strategies, Program Correctness, Program Verification & validation, Testing Automation &Testing Tools, Concept of Software Quality, Software Quality Attributes, Software Quality Metricsand Indicators, The SEI Capability Maturity Model CMM), SQA Activities, Formal SQA Approaches:Proof of correctness, Statistical quality assurance, Cleanroom process.

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MODULE III

Project Management and Project Management Tools

Software Configuration Management: Software Configuration Items and tasks, Baselines, Plan for Change, Change Control, Change Requests Management, Version Control, Risk Management: Risks and risk types, Risk Breakdown Structure (RBS), Risk Management Process: Risk identification, Risk analysis, Risk planning, Risk monitoring, Cost Benefit Analysis, Software Project Management Tools: CASE Tools, Planning and Scheduling Tools, MS-Project.

Text Books:

1. M. Cotterell, Software Project Management, Tata McGraw-Hill Publication.
2. Royce, Software Project Management, Pearson Education
4. Kieron Conway, Software Project Management, Dreamtech Press
5. S. A. Kelkar, Software Project Management, PHI Publication.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Understand the process of Software Project Management.
CO2	Conduct project planning activities that accurately forecast project costs.
CO3	Handle tools like MS Project & SVN
CO4	Understand the skills required for managing projects, project teams, and stakeholders
CO5	Understand the process of Software Project Management.